

TARUN RAVIKUMAR

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science

May 2026

- Relevant Coursework: Design & Analysis of Algorithms, Artificial Intelligence, Machine Learning, Computer Vision, Game AI, Data Structures, Computer Organization, Mobile & Ubiquitous Computing, User Interface Design, Systems Programming

EXPERIENCE

Software Engineer

Jul 2026 –

Oracle Cloud Infrastructure

Seattle, WA

Software Engineer Intern

May 2025 – Aug 2025

Oracle

Redwood City, CA

- Developed Oracle Cloud Navigator backend using FastAPI for cloud resource management and monitoring
- Implemented OpenSearch integration into Oracle Cloud Navigator app with advanced search capabilities, real-time filtering, and intelligent query optimization to replace legacy database calls and improve search performance by 40%
- Deployed and managed microservices using Kubernetes for container orchestration and scalable infrastructure
- Enhanced React/TypeScript frontend with state management and real-time data visualization for business requirements

Software Developer

Jan 2025 – Apr 2025

PopSign

Atlanta, GA

- Developed ASL vocab mobile app in collaboration with NTID and Google for gameplay and sign language recognition
- Ported ASL Engine games to Unity platform, ensuring smooth functionality transition and enhanced user experience
- Enhanced ASL datasets to improve recognition accuracy and support comprehensive game development pipeline

Undergraduate Student Researcher

Aug 2024 – Dec 2025

Georgia Tech College of Computing

Atlanta, GA

- Contributed to Automotive LiDAR Team, specializing in predictive modeling and sensor fusion for vehicle perception
- Developed ROS code for LeoRover using OpenCV for real-time data analysis and autonomous navigation control
- Refined LeoRover simulation in Gazebo by debugging ROS nodes to enhance robot performance in environment

Software Engineer Intern

May 2024 – Aug 2024

Mercedes-Benz

Sandy Springs, GA

- Developed service platform by expanding backend services and integrating NPS for customer analytics
- Automated SIOUX export data scraping using Python and Selenium, enhancing operational efficiency and accuracy
- Created VBA scripts to automate vulnerability tracking and financial analysis, reducing manual tasks by 30%

Undergraduate Student Researcher

Aug 2023 – Dec 2023

Penn State EECS

State College, PA

- Researched fuzzers, specifically BLEEM, for software testing and vulnerability discovery in embedded systems
- Conducted comparative performance analysis of BLEEM against traditional protocol fuzzers for systems testing

PROJECTS

tochi.one | Next.js, PyTorch, FastAPI, Twilio, Gemini API, shadcn/ui, Vercel

Mar 2025 – Present

- Reduced tenant defaults by 25% through PyTorch logistic regression risk assessment system for property screening
- Built responsive interface using Next.js and shadcn/ui, deployed on Vercel with continuous stakeholder engagement

AdBlockIRL | YOLOv8, OpenCV, Roboflow, Meta Quest AR

May 2025

- Trained YOLOv8 model via transfer learning from COCO weights with custom data augmentation (mosaic, mixup) and hyperparameter tuning, achieving 5ms latency detection of 100+ concurrent advertisements in live video streams
- Implemented real-time video manipulation using OpenCV with mosaic censoring effects on detected advertisements
- Developed for Meta Quest VR headset to censor advertisements in real-time during live video viewing
- Won \$2000 in prizes at Bain Capital Ventures AI Hackathon in NYC (FLORA track winner, Pond prize winner)

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, C, SQL, ROS, HTML/CSS, ARM Assembly, VBA

Frameworks & Libraries: PyTorch, TensorFlow.js, YOLOv8, OpenCV, Pandas, Scikit, Matplotlib, React.js, Next.js, Node.js,

Express.js, FastAPI, Flask, Django, Twilio, opensearch-py, aiobreaker, Selenium, Webpack, OAuth, React Native, Tailwind CSS, ROS

Tools & Platforms: Docker, Kubernetes, Git, Elasticsearch, MongoDB, NumPy, Huggingface, Vercel, Unity, Android Studio, Jira

Methodologies: Agile Development, Scrum, DevOps, CI/CD, Microservices Architecture, MLOps, A/B Testing